AI Study Guide: Foundations of Machine Learning

Page 1 | Chapter 1: Introduction to Machine Learning Paradigms

Machine Learning (ML) is a branch of artificial intelligence focused on building algorithms that learn patterns directly from training data to make predictions without explicit rules-based programming.

1. **Supervised Learning:** The algorithm learns from labeled pairs (features X, ground truth Y). Key tasks include classification (predicting discrete classes, e.g., spam vs ham) and regression (predicting continuous numeric values, e.g., housing prices). Common models: Linear Regression, Logistic Regression, Random Forests, and Support Vector Machines.
2. **Unsupervised Learning:** Training occurs on unlabeled data (features X only) to discover underlying cluster structures or reduce dimensions. Common algorithms include K-Means clustering, Hierarchical clustering, and Principal Component Analysis (PCA).
3. **Reinforcement Learning (RL):** An agent learns by taking actions within an environment to maximize cumulative reward signals. Central concepts include policy gradient, Q-learning, exploration vs exploitation trade-offs, and Markov Decision Processes (MDP).

Gradient Descent and Optimization: Machine learning models minimize an objective loss function L(theta). Weights are updated iteratively using the gradient of the loss with respect to parameters: theta = theta - alpha * grad(L), where alpha is the learning rate parameter. Stochastic Gradient Descent (SGD) and Adam optimizer are the most widely used optimizers.

AI Study Guide: Deep Learning & Transformers

Page 2 | Chapter 2: Deep Neural Networks and Self-Attention

Deep Learning utilizes layered artificial neural networks capable of learning hierarchical feature representations.

- **Artificial Neural Networks (ANN):** Composed of input layers, hidden layers with non-linear activation functions (ReLU, GELU, Sigmoid), and output layers. Backpropagation applies the calculus chain rule backwards across network layers to compute error gradients.
- **Convolutional Neural Networks (CNN):** Specialized for spatial grid data such as computer vision and medical imaging. They utilize convolution kernels for translation invariance and spatial pooling layers to extract edge and shape hierarchies.

The Transformer Architecture (Vaswani et al., 2017): Replaced recurrent recurrent architectures (RNN/LSTM) by using Self-Attention. The Attention formula computes attention weights across tokens: Attention(Q, K, V) = softmax((Q * K^T) / sqrt(d_k)) * V, where Q (Query), K (Key), and V (Value) are linear projections of token embeddings.

Multi-Head Attention: Allows the model to jointly attend to information from different representation subspaces at different positions. Modern Large Language Models like Google Gemini, GPT, and Claude rely on stacked transformer decoder or encoder-decoder blocks.

AI Study Guide: Retrieval-Augmented Generation (RAG)

Page 3 | Chapter 3: RAG Architecture & Vector Search

Retrieval-Augmented Generation (RAG) is an architectural pattern that enhances Large Language Models by retrieving authoritative knowledge from external documents (such as textbooks, research PDFs, or internal notes) before answering queries.

1. **Document Ingestion and Chunking:** Large PDFs are split into digestible text segments with overlapping boundaries (e.g., 800 characters with 150 overlap). This preserves complete sentences and context across page boundaries.

2. **Semantic Vector Embeddings:** Neural embedding models (such as text-embedding-004) convert text chunks into dense numeric vectors in high-dimensional space. Semantically similar passages will lie close together regardless of exact keyword phrasing.

3. **Vector Indexing and Similarity Search:** Vector stores calculate distance metrics, predominantly Cosine Similarity: Cosine_Sim(A, B) = (A . B) / (||A|| * ||B||). Top-K nearest chunks are retrieved for any incoming student question.

4. Grounded Prompt Synthesis: The retrieved excerpts are dynamically injected into the system context prompt along with page citations. The LLM is instructed to answer strictly based on the retrieved excerpts, drastically eliminating hallucinations.

Key Study Takeaways for RAG: RAG decouples model training from knowledge updates, enables instant citation verification to source page numbers, and keeps private academic documents secure without expensive fine-tuning.